

An Energy-Aware, Contact-Graph Routing Optimization Strategy for Heterogeneous Deep Space SmallSat Relays

Jason Pandian

Abstract—Contact Graph Routing (CGR) is the de facto standard for store-carry-forward routing in deep-space Delay-Tolerant Networks (DTN). While CGR leverages the deterministic predictability of orbital contact schedules, it implicitly assumes that relay nodes possess sufficient energy and storage to execute all scheduled transmissions. This assumption breaks down in heterogeneous networks containing resource-constrained SmallSats, where standard CGR drains relay batteries to critical levels, causing packet drops. State-of-the-Art Energy-Aware CGR (ECGR) protects relay health but suffers from resource conservatism due to using instantaneous state metrics, yielding lower delivery ratios. This paper proposes Predictive Energy and Capacity-Aware Contact Graph Routing (P-ECGR), which introduces a future energy prediction model and a reservation-based energy booking registry. By tracking committed transmissions, P-ECGR prevents energy overbooking and load-balances traffic across heterogeneous relays. We evaluate P-ECGR via discrete-event simulation of a Mars relay network using contact plans from synthetic SPICE ephemeris data. Over 10 Monte Carlo runs, P-ECGR maintains zero packet drops and keeps the SmallSat battery safe (spending $2.0 \pm 4.7\%$ of the time below 20% SoC, compared to $8.5 \pm 13.2\%$ for CGR and $1.7 \pm 4.6\%$ for SOTA ECGR), while increasing the bundle delivery ratio to $90.0 \pm 1.7\%$ (outperforming CGR's $83.7 \pm 8.9\%$ and SOTA ECGR's $88.9 \pm 1.9\%$), and reducing latency compared to both standard CGR and SOTA ECGR. The proposed algorithm adds negligible computational overhead, making it suitable for flight-grade onboard processors.

Index Terms—Delay-tolerant networking, contact graph routing, deep space communications, SmallSat, energy-aware routing, Mars relay network

I. Introduction

A. Deep Space Communication Bottlenecks

Modern space exploration is transitioning from isolated, single-spacecraft missions to complex, multi-agent networks comprising rovers, landers, orbiters, and small satellite constellations. This paradigm shift requires high-bandwidth, reliable communication architectures. However, deep-space communications operate in environments characterized by extreme physical constraints, such as planetary occultations, orbital eclipses, massive free-space path losses over interplanetary distances, and solar plasma interference.

Under these conditions, standard terrestrial network protocols, such as TCP/IP, fail catastrophically. TCP

relies on continuous end-to-end paths and assumes that round-trip times (RTT) are short and packet loss is primarily caused by congestion. In contrast, deep-space links involve one-way light times (OWLT) ranging from several minutes to hours, and connectivity is highly intermittent due to orbital mechanics. To address these challenges, the Space Internet architecture leverages Delay-Tolerant Networking (DTN) [1], [2] and the Bundle Protocol version 7 (BPv7) [3]. DTN implements a store-carry-forward routing paradigm, where intermediate nodes store data in non-volatile memory buffers until a downstream communication opportunity (contact) becomes available.

B. Store-Carry-Forward and CGR

In space missions, the motion of celestial bodies and spacecraft is governed by orbital mechanics, rendering communication opportunities highly predictable. By utilizing ephemeris data, space agencies pre-compute a contact plan that lists every scheduled communication window between nodes, including start times, durations, data rates, and propagation delays.

Contact Graph Routing (CGR) [4] is the standard algorithm designed for these planned, scheduled-contact topologies. CGR represents the contact plan as a time-varying directed graph (the contact graph), where vertices represent communication contacts and edges denote time-ordered transitions between contacts. When a node needs to route a data packet (bundle), CGR runs a modified version of Dijkstra's algorithm to compute the path that minimizes end-to-end delivery delay (earliest arrival time). This approach is highly effective in homogeneous networks composed of well-provisioned spacecraft, such as NASA's Mars Reconnaissance Orbiter (MRO) or Mars Odyssey, which possess large solar arrays and high-capacity battery systems.

C. Heterogeneous Space Networks and SmallSats

The emergence of SmallSats and CubeSats (e.g., 3U and 6U form factors) has enabled low-cost, distributed missions, such as the MarCO CubeSats that tracked the InSight lander. Space agencies are planning to deploy swarms of SmallSats to provide localized relay coverage around Mars and the Moon [5], [6]. However, these micro-platforms have extremely tight Size, Weight, and Power (SWaP) constraints.

A typical 3U SmallSat relay possesses a small solar panel area, restricting average power generation to less than 10W. Its battery capacity is restricted (e.g., 15–30Wh), and its onboard buffer storage is limited (e.g., 256–512MB). Crucially, operating a high-power radio transceiver during a transmission contact can consume 15–25W, which is far greater than the average solar charging rate. As a result, active communication rapidly drains the satellite’s battery. If the SmallSat is in Mars’ shadow (orbital eclipse), it cannot generate solar power, and transmission must rely solely on the battery.

D. Problem Statement

Standard CGR is resource-blind. It assumes that if a contact window is listed in the contact plan, the transmitting node has sufficient electrical energy and storage capacity to complete the transfer. In a heterogeneous network consisting of a rover, a constrained SmallSat relay, a flagship orbiter, and Earth ground stations, CGR will route massive scientific images through the SmallSat relay simply because it offers the earliest contact window. This decision can deplete the SmallSat’s battery, causing a low-voltage shutdown that drops all queued bundles and suspends housekeeping telemetry, jeopardizing the spacecraft’s safety.

To protect relay node health, State-of-the-Art (SOTA) Energy-Aware CGR (ECGR) algorithms evaluate node resource states during route computation. However, SOTA ECGR relies on instantaneous battery state-of-charge (SoC) metrics at the routing epoch. This approach has two main limitations:

- 1) Resource Conservatism: It penalizes routes through a node with a low battery state at t_{now} , even if that node will spend the next three hours in full sunlight charging its battery before the transmission contact actually begins. This conservatism reduces the delivery ratio and increases latency.
- 2) Energy Overbooking: In multi-bundle scenarios, a source node processes bundles sequentially. If the source evaluates routes for twenty bundles in the same time step, each computation sees a safe instantaneous SoC at the relay. Consequently, the source routes all twenty bundles through the same SmallSat. The cumulative energy required to transmit this data far exceeds the SmallSat’s battery capacity, causing battery exhaustion and packet drops.

E. Proposed Solution and Contributions

To overcome these limitations, this paper proposes Predictive Energy and Capacity-Aware Contact Graph Routing (P-ECGR). P-ECGR integrates a future orbital solar-charging prediction model with a local reservation-based energy booking registry. The prediction model projects the relay’s battery state forward to the start time of the future contact, taking into account predicted solar generation and eclipse periods. The energy-booking registry tracks

committed transmissions, preventing energy overbooking. The contributions of this work are summarized as follows:

- We identify and formulate the energy overbooking bottleneck in state-of-the-art energy-aware routing for heterogeneous space networks. We propose P-ECGR, a routing algorithm that incorporates future solar-charging predictions and a local energy booking registry to prevent energy overbooking and balance traffic load.
- We present a modular Python simulation framework that implements circular Keplerian orbit propagation, eclipse checks, and resource-constrained node models.
- We evaluate CGR, SOTA ECGR, and the proposed P-ECGR on a Mars relay scenario, demonstrating that P-ECGR achieves higher delivery ratios and lower latencies than SOTA ECGR while maintaining zero packet drops.

F. Paper Outline

The remainder of this paper is structured as follows. Section II reviews related work in DTN and resource-aware routing. Section III defines the system model, including topology, orbits, battery dynamics, and buffers. Section IV details the routing formulations, presenting the algorithms for CGR, SOTA ECGR, and proposed P-ECGR. Section V describes the simulation setup and parameters. Section VI analyzes the results, and Section VII concludes the paper.

II. Related Work

A. Delay-Tolerant Networking and Bundle Protocol

The concept of Delay-Tolerant Networking originated from efforts to design an Interplanetary Internet, led by Cerf and others under the DARPA-funded Interplanetary Internet project. Fall [1] proposed a generalization of these concepts for challenged terrestrial and extraterrestrial environments, introducing the store-carry-forward routing paradigm. This architecture was formalized by the DTN Research Group (DTNRG) in RFC 4838 [2].

The Bundle Protocol (BP) serves as the primary protocol layer for DTN. BP resides between the application and transport/convergence layers, encapsulating application data into protocol data units called bundles. The protocol has evolved from version 6 (BPv6) to the modern version 7 (BPv7), which was standardized in RFC 9171 [3] to support security and extensibility blocks. Satellite networking reviews, such as those by Caini et al. [7], have demonstrated the benefits of BP in mitigating the effects of long round-trip times and frequent link disconnections in space communication links.

B. Contact Graph Routing Advancements

Contact Graph Routing was developed by Burleigh [4] to exploit the scheduled, predictable nature of space networks. Unlike routing protocols in terrestrial networks that rely on periodic route discovery packets, CGR utilizes pre-distributed contact plans. Fraire et al. [8] provided a

tutorial on CGR, detailing its Dijkstra-based path finding mechanics.

Several enhancements have been proposed to improve CGR's performance. Araniti et al. [9] surveyed various routing strategies, including Space Active Bundle Routing (SABR) and opportunistic link integration. Seguí et al. [10] adapted CGR for Mars relay operations planning, demonstrating the feasibility of using orbital schedules to route data from surface landers to Earth ground stations. However, these classical implementations assume that relay nodes have unlimited energy and memory buffers, an assumption that is invalid for SmallSat relays.

C. Energy-Aware and Congestion-Aware Routing

Congestion management in DTN space networks has been studied extensively. Madoery et al. [11] proposed queue management and congestion-avoidance routing techniques to prevent buffer overflows at satellite relays. While these techniques protect storage buffers, they do not address battery limits.

Energy-constrained routing has received less attention in the context of CGR. Alessi et al. [12] proposed incorporating energy metrics into CGR, penalizing routes through nodes with low battery states. However, their formulation evaluated only the current, instantaneous battery SoC at the routing epoch. This approach leads to resource conservatism (avoiding nodes that will recharge before transmission) and fails to address the energy overbooking problem where multiple bundles are routed through the same node simultaneously.

Machine learning and reinforcement learning approaches have been proposed for DTN routing (e.g., Dudukovich et al. [13]). However, these algorithms require significant computational resources, making them unsuitable for radiation-hardened spacecraft processors, such as the RAD750 (operating at 133–200 MHz) or LEON3 processors, which have strict memory and CPU utilization limits. In contrast, our proposed P-ECGR algorithm has low computational overhead and is backward-compatible with existing CGR systems.

III. System Model

A. Network Topology and Scenario

We consider a four-node deep-space relay network modeling a realistic Mars exploration scenario, illustrated in Fig. 1. The network consists of:

- Mars Surface Rover (Node 0): Located at Jezero Crater (4.5°S, 77.4°E). It acts as the primary data source, generating scientific images, critical telemetry, and housekeeping data. The rover is powered by a Multi-Mission Radioisotope Thermoelectric Generator (MMRTG) providing constant power generation.
- SmallSat Relay (Node 1): A resource-constrained 3U CubeSat in a near-polar Sun-synchronous circular orbit. It has a small battery, constrained buffer, and is solar-powered, meaning its power generation is zero during eclipses.

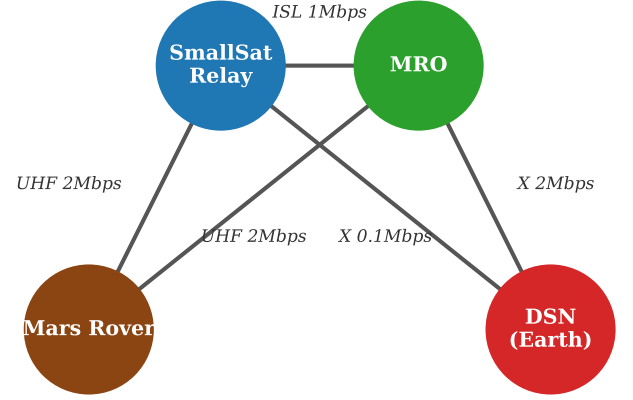


Fig. 1: Mars relay network topology showing the four-node scenario with link types and data rates. The SmallSat relay (Node 1) is the resource-constrained bottleneck.

- Mars Reconnaissance Orbiter (Node 2): A flagship-class orbiter in low Mars orbit. It possesses a high-capacity battery, large solar arrays, and a large storage buffer, allowing it to act as a high-capacity relay.
- Deep Space Network (Node 3): Earth ground stations, representing the destination. The DSN has effectively unlimited energy, buffer capacity, and processing power.

B. Orbits and Visibility Model

The positions of the SmallSat relay and MRO are propagated using Keplerian circular orbits around Mars. The orbital radius r is defined as:

$$r = R_M + h \quad (1)$$

where $R_M = 3389.5$ km is the mean radius of Mars, and h is the altitude of the satellite (400 km for SmallSat, 300 km for MRO). The orbital period T is computed using Kepler's third law:

$$T = 2\pi \sqrt{\frac{r^3}{GM_M}} \quad (2)$$

where $GM_M = 4.282837 \times 10^{13} \text{ m}^3/\text{s}^2$ is the gravitational parameter of Mars. This yields an orbital period of approximately 7,560 s (2.1 h) for the SmallSat and 6,840 s (1.9 h) for the MRO.

A communication link between a surface node (rover) and an orbiter is feasible only when the orbiter is above the rover's local horizon. Geometrically, the elevation angle θ_{el} of the satellite relative to the rover must exceed a minimum elevation mask $\theta_{mask} = 10^\circ$ to clear topography and minimize atmospheric attenuation. The line-of-sight

vector $\mathbf{d}$ from the rover position $\mathbf{p}_r$ to the satellite position $\mathbf{p}_s$ is:

$$\mathbf{d} = \mathbf{p}_s - \mathbf{p}_r \quad (3)$$

The elevation angle is defined as:

$$\theta_{el} = 90^\circ - \arccos\left(\frac{\mathbf{d} \cdot \mathbf{p}_r}{\|\mathbf{d}\| \|\mathbf{p}_r\|}\right) \quad (4)$$

A contact is created when $\theta_{el} \geq \theta_{mask}$ and the line of sight is not obstructed by Mars.

C. Energy Model

For each non-ground node, the battery energy state $E(t)$ (Wh) is modeled as a continuous state-of-charge equation:

$$E(t + \Delta t) = \min(E_{\max}, \max(0, E(t) + \Delta E_{\text{net}})) \quad (5)$$

where $E_{\max}$ is the battery capacity, and ΔE_{net} is the net energy change over time step Δt , computed as:

$$\Delta E_{\text{net}} = (P_{\text{gen}}(t) - P_{\text{cons}}(t)) \cdot \frac{\Delta t}{3600} \quad [\text{Wh}] \quad (6)$$

The power generation $P_{\text{gen}}(t)$ depends on whether the node is in Mars' shadow:

$$P_{\text{gen}}(t) = \begin{cases} 0.0 & \text{if in eclipse} \\ P_{\text{solar}} & \text{otherwise} \end{cases} \quad (7)$$

Eclipses occur when Mars obstructs the line of sight between the satellite and the Sun. The power consumption $P_{\text{cons}}(t)$ is:

$$P_{\text{cons}}(t) = P_{\text{idle}} + P_{\text{tx}} \cdot \mathbb{I}_{\text{tx}}(t) + P_{\text{rx}} \cdot \mathbb{I}_{\text{rx}}(t) \quad (8)$$

where P_{idle} is the baseline system power, P_{tx} is the additional power consumed by the radio transmitter during active transmission, P_{rx} is the power consumed by the receiver during active reception, and $\mathbb{I}_{\text{tx}}, \mathbb{I}_{\text{rx}} \in \{0, 1\}$ are indicators of transmission and reception activity.

The state of charge (SoC) is defined as:

$$\text{SoC}(t) = \frac{E(t)}{E_{\max}} \quad (9)$$

If the battery SoC drops to 0.0, the node shut downs, dropping all bundles currently stored in its buffer.

D. Buffer Model

Each node n has a physical buffer storage capacity $B_{\max}^n$ bytes. The buffer utilization $U_B(t)$ is the fraction of capacity occupied by queued bundles:

$$U_B(t) = \frac{\sum_{b \in \mathcal{Q}(t)} |b|}{B_{\max}^n} \quad (10)$$

where $\mathcal{Q}(t)$ is the set of bundles stored in node n 's queue at time t , and $|b|$ is the size of bundle b in bytes. The available buffer space is:

$$B_{\text{avail}}^n(t) = B_{\max}^n - \sum_{b \in \mathcal{Q}(t)} |b| \quad (11)$$

If a node receives a bundle b when $B_{\text{avail}}^n < |b|$, the bundle is dropped due to buffer overflow.

E. Traffic Generation Model

The rover generates three classes of traffic, characterized by priority, bundle size, and arrival intervals:

- 1) Priority 1 (Critical Telemetry): Consists of small health and status reports (size range 1.0–5.0 MB) generated at short intervals (1000–2000 s). These bundles require low latency.
- 2) Priority 2 (Scientific Images): Consists of large scientific files (size range 15.0–90.0 MB) generated at longer intervals (2000–4000 s). These bundles represent the bulk of the network volume.
- 3) Priority 3 (Housekeeping Data): Consists of small, routine reports (size range 0.2–2.0 MB) generated at short intervals (500–1000 s). These have the lowest priority.

IV. Routing Algorithms and Formulations

In this section, we present the formulations of the three routing algorithms evaluated in this study: standard Contact Graph Routing (CGR), SOTA Energy-Aware CGR (ECGR), and our proposed Predictive Energy and Capacity-Aware CGR (P-ECGR).

A. Standard Contact Graph Routing (CGR)

Standard CGR aims to minimize delivery delay. Given a bundle b created at time t_{now} , CGR identifies all feasible routes to the destination. For each route r , composed of a sequence of contacts $r = (c_1, c_2, \dots, c_k)$, the arrival time $T_{\text{arr}}(r)$ is:

$$T_{\text{arr}}(r) = t_{\text{end}}(c_k) + \tau_{\text{owl}}(c_k) \quad (12)$$

CGR filters out routes that violate link capacities or create routing loops. A route r is feasible if:

$$C_{\text{resid}}(c_i) \geq |b| \quad \forall c_i \in r \quad (13)$$

where $C_{\text{resid}}(c_i)$ is the residual capacity of contact c_i . The optimal route r^* is selected to minimize arrival time:

$$r^* = \arg \min_{r \in \mathcal{R}_f} T_{\text{arr}}(r) \quad (14)$$

where $\mathcal{R}_f$ is the set of capacity-feasible routes. CGR is resource-blind: it does not evaluate whether the nodes along the route have the energy to complete the transmissions.

B. State-of-the-Art Energy-Aware CGR (ECGR)

SOTA ECGR incorporates relay battery and buffer states into a multi-attribute cost function:

$$\mathcal{C}_{\text{ECGR}}(r, b) = \alpha \cdot \hat{D}(r) + \beta \cdot \Phi_E(r) + \gamma \cdot \Phi_B(r, b) \quad (15)$$

where:

- $\hat{D}(r) = D(r)/D_{\max}$ is the normalized route delay, with $D(r) = T_{\text{arr}}(r) - t_{\text{now}}$ and $D_{\max} = \max_{r' \in \mathcal{R}_f} D(r')$.
- $\Phi_E(r) = 1.0 / \min_{n \in \mathcal{R}(r)} \text{SoC}_n(t_{\text{now}})$ is the energy penalty, which is inversely proportional to the minimum instantaneous SoC among the relay nodes $\mathcal{R}(r)$.

- $\Phi_B(r, b) = \max_{n \in \mathcal{R}(r)} (|b|/B_{\text{avail}}^n(t_{\text{now}}))$ is the buffer penalty, reflecting the fraction of available storage the bundle would occupy.
- α, β, γ are dynamic weights scaled based on bundle priority and instantaneous warning/critical thresholds.

During route selection, ECGR filters out routes if any relay node cannot transmit the bundle based on its current energy state:

$$E_n(t_{\text{now}}) < (P_{\text{tx}}^n + P_{\text{idle}}^n) \cdot \frac{8 \cdot |b|}{R_{\text{link}}} \cdot \frac{1}{3600} \quad (16)$$

The SOTA ECGR route-selection procedure is detailed in Algorithm 1.

Algorithm 1 SOTA ECGR Route Selection

Require: Contact plan $\mathcal{C}$, bundle b , node states $\mathcal{N}$, current time t_{now}

```

Ensure: Selected route  $r^*$  or  $\emptyset$ 
1:  $\mathcal{R} \leftarrow \text{FindAllRoutes}(\mathcal{C}, b.\text{src}, b.\text{dst}, t_{\text{now}})$ 
2:  $\mathcal{R}_f \leftarrow \emptyset$ 
3: for each  $r \in \mathcal{R}$  do
4:    $\text{capacity\_ok} \leftarrow \text{True}$ 
5:   for each hop  $h \in r.\text{hops}$  do
6:     if  $h.\text{residual\_capacity} < b.\text{size\_bytes}$  then
7:        $\text{capacity\_ok} \leftarrow \text{False}$ 
8:     end if
9:   end for
10:  if not  $\text{capacity\_ok}$  then
11:    continue
12:  end if
13:   $\text{relay\_ok} \leftarrow \text{True}$ 
14:  for each relay node  $n \in r.\text{relay\_nodes}$  do
15:     $\text{tx\_time} \leftarrow b.\text{size\_bytes} \cdot 8/r.\text{bottleneck\_rate}$ 
16:    if not  $\mathcal{N}[n].\text{can\_store}(b.\text{size\_bytes})$  or not  $\mathcal{N}[n].\text{can\_transmit}(\text{tx\_time})$  then
17:       $\text{relay\_ok} \leftarrow \text{False}$ 
18:    end if
19:  end for
20:  if  $\text{relay\_ok}$  then
21:     $\mathcal{R}_f \leftarrow \mathcal{R}_f \cup \{r\}$ 
22:  end if
23: end for
24: if  $\mathcal{R}_f = \emptyset$  then
25:   return  $\emptyset$ 
26: end if
27:  $D_{\text{max}} \leftarrow \max_{r \in \mathcal{R}_f} D(r)$ 
28: Selected route  $r^* \leftarrow \arg \min_{r \in \mathcal{R}_f} \mathcal{C}_{\text{ECGR}}(r, b; \mathcal{N}, D_{\text{max}}, t_{\text{now}})$ 
29: return  $r^*$ 

```

C. Proposed Predictive and Energy-Booked ECGR (P-ECGR)

Proposed P-ECGR replaces the instantaneous metrics of SOTA ECGR with a predictive orbital energy model and a local energy booking registry to resolve the energy overbooking problem.

1) Future Energy Prediction Model: Instead of evaluating the instantaneous energy state $E_n(t_{\text{now}})$, P-ECGR predicts the battery energy state $E_{\text{pred}}(n, t_{\text{start}})$ at the future start time t_{start} of the contact window. The prediction is computed by integrating the nominal solar generation and idle drain rates over the propagation delay:

$$E_{\text{pred}}(n, t_{\text{start}}) = E_n(t_{\text{now}}) + \int_{t_{\text{now}}}^{t_{\text{start}}} (P_{\text{gen}}(n, t) - P_{\text{idle}}(n)) dt \quad (17)$$

where $P_{\text{gen}}(n, t)$ is modeled as $P_{\text{solar}}^n \cdot (1 - \text{shadow}(n, t))$, and $\text{shadow}(n, t) \in \{0, 1\}$ is a step function derived from predicted orbital eclipse windows. The predicted energy is bounded by the battery capacity E_{max}^n :

$$0 \leq E_{\text{pred}}(n, t_{\text{start}}) \leq E_{\text{max}}^n \quad (18)$$

2) Energy-Booking Reservation Registry: To prevent energy overbooking, P-ECGR maintains a local reservation registry $\mathcal{R}_{\text{book}}$. When a route r is selected for bundle b , the expected transmission energy $E_{\text{tx}}(b)$ is computed as:

$$E_{\text{tx}}(b) = (P_{\text{tx}}(n) + P_{\text{idle}}(n)) \cdot \frac{8 \cdot |b|}{R_{\text{link}}} \cdot \frac{1}{3600} \quad [\text{Wh}] \quad (19)$$

This energy consumption is booked in the registry for node n at the contact's start time t_{start} .

When evaluating subsequent bundles, the net predicted energy $E_{\text{net}}(n, t_{\text{start}})$ is computed by subtracting all booked energy allocations scheduled to transmit prior to t_{start} :

$$E_{\text{net}}(n, t_{\text{start}}) = E_{\text{pred}}(n, t_{\text{start}}) - \sum_{k \in \mathcal{K}_n(t_{\text{now}}, t_{\text{start}})} E_{\text{tx}}(b_k) \quad (20)$$

where $\mathcal{K}_n(t_{\text{now}}, t_{\text{start}})$ is the set of bundle IDs booked for transmission on node n during the interval $[t_{\text{now}}, t_{\text{start}}]$. Registry entries scheduled in the past ($t < t_{\text{now}}$) are pruned, as their actual energy drain is already reflected in the node's real-time state $E_n(t_{\text{now}})$.

3) Energy Penalty deadband Function: Let the predicted state of charge be $\text{SoC}_{\text{pred}}(n) = E_{\text{net}}(n, t_{\text{start}})/E_{\text{max}}^n$. The predicted post-transmission SoC after sending bundle b is:

$$\text{SoC}_{\text{post}}(n) = \text{SoC}_{\text{pred}}(n) - \frac{E_{\text{tx}}(b)}{E_{\text{max}}^n} \quad (21)$$

To avoid penalizing routes when the relay operates in the safe energy zone, we implement a deadband penalty function $\Phi_E(r, b) = \sum_{n \in \mathcal{R}(r)} \phi_E(n, b)$, where:

$$\phi_E(n, b) = \begin{cases} 0.0 & \text{SoC}_{\text{post}}(n) \geq \theta_{\text{warn}} \\ \lambda \cdot \frac{\theta_{\text{warn}} - \text{SoC}_{\text{post}}(n)}{\theta_{\text{warn}}} & \theta_{\text{crit}} \leq \text{SoC}_{\text{post}}(n) < \theta_{\text{warn}} \\ \Omega & \text{SoC}_{\text{post}}(n) < \theta_{\text{crit}} \end{cases} \quad (22)$$

where $\theta_{\text{warn}} = 0.35$ and $\theta_{\text{crit}} = 0.20$ are the warning and critical battery thresholds, $\lambda = 10.0$ is the warning scale factor, and $\Omega = 50.0$ is the severe critical penalty.

4) Priority-Based Weight Adaptation: The weights (α, β, γ) scale adaptively to balance delay minimization with resource protection. First, the base weights $(\alpha_0, \beta_0, \gamma_0)_p$ are determined by the bundle's priority class p :

$$(\alpha_0, \beta_0, \gamma_0)_p = \begin{cases} (2.0, 0.5, 0.5) & p = 1 \text{ (Critical)} \\ (1.0, 1.0, 1.0) & p = 2 \text{ (Normal)} \\ (0.3, 2.0, 2.0) & p = 3 \text{ (Low)} \end{cases} \quad (23)$$

Second, the weights are scaled dynamically according to the net predicted energy state:

$$\beta' = \beta_0 \cdot \begin{cases} 4.0 & \text{if } E_{\text{net}} < \theta_{\text{crit}} \cdot E_{\text{max}} \\ 2.0 & \text{if } \theta_{\text{crit}} \cdot E_{\text{max}} \leq E_{\text{net}} < \theta_{\text{warn}} \cdot E_{\text{max}} \\ 1.0 & \text{otherwise} \end{cases} \quad (24)$$

$$\gamma' = \gamma_0 \cdot \begin{cases} 3.0 & \text{if } U_B^{\text{max}} > 0.70 \\ 1.0 & \text{otherwise} \end{cases} \quad (25)$$

5) Feasibility Filtering: Before cost calculation, candidate routes are filtered for feasibility. A route r is feasible for bundle b if and only if:

- 1) The residual capacity of each contact hop exceeds $|b|$:

$$C_{\text{resid}}(c) \geq |b| \quad \forall c \in r \quad (26)$$

- 2) The available buffer at each relay node exceeds $|b|$:

$$B_{\text{avail}}^n \geq |b| \quad \forall n \in \mathcal{R}(r) \quad (27)$$

- 3) The net predicted energy at transmission start time exceeds the energy required for transmission:

$$E_{\text{net}}(n, t_{\text{start}}) \geq E_{\text{tx}}(b) \quad \forall n \in \mathcal{R}(r) \quad (28)$$

The complete P-ECGR route-selection procedure is detailed in Algorithm 2.

V. Python Simulation Setup

To evaluate the performance of CGR, SOTA ECGR, and proposed P-ECGR, we developed a discrete-time event-driven Python simulator representing the deep-space Mars relay scenario.

A. Simulator Package Structure

The simulator codebase is structured as a modular package containing the following core files:

- `config.py`: Defines all global parameters, including simulation duration (86,400s), time step size (10s), node SWaP specs (Table I), link parameters, and priority weights.
- `models.py`: Implements the data models for 'Contact', 'Bundle', 'Node', and 'Route'. The 'Node' class encapsulates energy update equations, battery charging/discharging logic, and buffer storage checks.
- `spice_data.py`: Implements the synthetic SPICE ephemeris engine, generating Keplerian orbital trajectories for the SmallSat relay and MRO around Mars

Algorithm 2 P-ECGR Route Selection and Energy Booking

Require: Contact plan $\mathcal{C}$, bundle b , node states $\mathcal{N}$, local booking registry $\mathcal{R}_{\text{book}}$, time t
 Ensure: Selected route r^* or $\emptyset$

- 1: $\mathcal{R} \leftarrow \text{FindAllRoutes}(\mathcal{C}, b.\text{src}, b.\text{dst}, t)$
- 2: $\mathcal{R}_f \leftarrow \emptyset$
- 3: $\text{CleanExpiredBookings}(\mathcal{R}_{\text{book}}, t)$
- 4: $\text{RemovePriorReservations}(\mathcal{R}_{\text{book}}, b.\text{id})$
- 5: for each $r \in \mathcal{R}$ do
- 6: if $\text{CheckLinkCapacity}(r, b)$ and $\text{CheckBufferSpace}(r, b, \mathcal{N})$ and $\text{CheckPredictiveEnergy}(r, b, \mathcal{N}, \mathcal{R}_{\text{book}}, t)$ then
- 7: $\mathcal{R}_f \leftarrow \mathcal{R}_f \cup \{r\}$
- 8: Compute predicted SoCs for r using $E_{\text{net}}(n, t_{\text{start}})$
- 9: end if
- 10: end for
- 11: if $\mathcal{R}_f = \emptyset$ then
- 12: return $\emptyset$
- 13: end if
- 14: $D_{\text{max}} \leftarrow \max_{r \in \mathcal{R}_f} D(r)$
- 15: $(\alpha, \beta, \gamma) \leftarrow \text{GetAdaptiveWeights}(b, \mathcal{N}, \mathcal{R}_{\text{book}}, t)$
- 16: $r^* \leftarrow \arg \min_{r \in \mathcal{R}_f} \mathcal{C}(r, b; \alpha, \beta, \gamma, D_{\text{max}})$
- 17: if $r^* \neq \emptyset$ then
- 18: for each hop $h \in r^*$ with source node $n \in \mathcal{R}(r^*)$ do
- 19: $E_{\text{tx}} \leftarrow \text{ComputeTxEnergy}(n, b, h.\text{rate})$
- 20: $\text{RegisterBooking}(\mathcal{R}_{\text{book}}, n, h.\text{start_time}, E_{\text{tx}}, b.\text{id})$
- 21: end for
- 22: end if
- 23: return r^*

and checking for eclipses and line-of-sight visibility windows.

- `contact_plan.py`: Generates contact plans based on the visibility calculations and exports them to JSON and standard JPL ION-DTN configuration format.
- `routing.py`: Implements the routing decision logic for 'CGRRouter', 'ECGRRouter', and 'EBECGRRouter' (P-ECGR).
- `simulator.py`: Implements the main discrete-event execution loop. In each step, the simulator updates node batteries, injects newly generated bundles, and evaluates route options for queued bundles.

B. Synthetic SPICE and Orbital Generation

The simulator generates orbital trajectories utilizing circular Keplerian motion. The position vector $\mathbf{r}(t)$ of a satellite in orbit is represented as:

$$\mathbf{r}(t) = r \cos(\omega t) \hat{\mathbf{u}} + r \sin(\omega t) \hat{\mathbf{v}} \quad (29)$$

where r is the orbital radius, $\omega = 2\pi/T$ is the angular velocity, and $\hat{\mathbf{u}}, \hat{\mathbf{v}}$ are orthogonal unit vectors defining the orbital plane. The inclination angles are set to 87° for the SmallSat relay and 93° for the MRO, representing typical polar science orbits. Eclipses are modeled using a

TABLE I: Simulation Parameters

Parameter	Value
Simulation Duration	86,400 s (24.0 h)
SmallSat Battery	16.0 Wh
SmallSat Solar Power	8.0 W (avg)
SmallSat Buffer	512 MB
SmallSat Tx Power	15.0 W
MRO Battery	1120.0 Wh
MRO Buffer	8192 MB
Rover-SmallSat Rate	2 Mbps (UHF)
MRO-DSN Rate	2 Mbps (X-band)
Earth-Mars OWLT	750 s
Monte Carlo Runs	10

cylindrical projection of Mars' shadow. A satellite is in eclipse if:

$$\mathbf{r}(t) \cdot \hat{\mathbf{s}} < 0 \quad \text{and} \quad \|\mathbf{r}(t) - (\mathbf{r}(t) \cdot \hat{\mathbf{s}})\hat{\mathbf{s}}\| < R_M \quad (30)$$

where $\hat{\mathbf{s}}$ is the unit vector pointing toward the Sun.

C. Simulation Scenario and Traffic Loading

The simulation models a 24-hour Martian day. The rover (Node 0) generates data according to the three profiles described in Section III. A total of 199 bundles are generated over the day, representing approximately 2.1 GB of scientific and telemetry data. The SmallSat relay (Node 1) and MRO (Node 2) act as relays to deliver these bundles to the DSN (Node 3) on Earth.

The link from MRO to DSN has a data rate of 2 Mbps, and MRO is well-provisioned (battery capacity 1120 Wh, solar power 1000 W). In contrast, the link from the SmallSat relay to DSN is slower (500 kbps), and the SmallSat is highly constrained (battery capacity 16 Wh, solar power 8 W, transmission power draw 15 W). This configuration creates a resource bottleneck at the SmallSat relay, testing the resource-awareness of the routing algorithms.

D. Simulation Parameters

The simulation parameters are summarized in Table I.

E. Monte Carlo Configuration

To obtain statistically significant results, we execute 10 independent Monte Carlo runs. The orbital contact plan is kept identical across all runs (since orbits are deterministic), but the traffic generation seeds are varied deterministically as $42 + 17i$ for run i . This approach introduces variance in bundle creation times, sizes, and priority sequences.

F. Performance Evaluation Metrics

To quantify the behavior of the three routing schemes (CGR, ECGR, and P-ECGR) we collect a set of complementary metrics that capture throughput, latency, reliability, and resource usage of the SmallSat relay. All metrics are computed over the 10 Monte-Carlo simulation runs and reported as mean $\pm$ one standard deviation.

- Bundle Delivery Ratio (BDR) (%):

$$\text{BDR} = \frac{N_{\text{delivered}}}{N_{\text{generated}}} \times 100, \quad (31)$$

where $N_{\text{generated}}$ is the total number of bundles injected into the network and $N_{\text{delivered}}$ is the number that reach the destination before the 24-hour simulation deadline. BDR directly measures the effectiveness of the routing algorithm in utilizing available contacts.

- Average End-to-End Latency (seconds):

$$\bar{L} = \frac{1}{N_{\text{delivered}}} \sum_{i=1}^{N_{\text{delivered}}} (t_i^{\text{arrival}} - t_i^{\text{inject}}), \quad (32)$$

where t_i^{inject} is the injection time of bundle i and t_i^{arrival} is the time it is received at the ground station. This metric captures the timeliness of data delivery, which is critical for time-sensitive telemetry and scientific payloads.

- 95th-percentile Latency (seconds): The 95th percentile of the latency distribution,

$$L_{95} = \text{percentile}_{0.95} \{t_i^{\text{arrival}} - t_i^{\text{inject}}\},$$

highlights worst-case delivery delays experienced by a small fraction of bundles, providing insight into tail-performance for high-priority traffic.

- Dropped Bundles (count): The number of bundles that are removed from the simulation due to either (i) energy exhaustion (the SmallSat's battery falls below the transmission threshold) or (ii) buffer overflow. This metric quantifies reliability loss resulting from resource constraints.
- SmallSat Minimum State-of-Charge (SoC) (%): The lowest battery SoC observed on the SmallSat during a run. It is a direct indicator of the routing algorithm's impact on relay health.
- Time Below Critical SoC (% of simulation time):

$$\text{Below}_{20\%} = \frac{\int_0^T \mathbf{1}(\text{SoC}(t) < 20\%) dt}{T} \times 100, \quad (33)$$

where T is the total simulation horizon (24 h). This metric captures how often the SmallSat operates in a vulnerable energy regime.

- Total Data Delivered (MB): The cumulative volume of payload data successfully transferred to the ground station, computed as the sum of the sizes of all delivered bundles. It reflects the overall scientific return of each routing strategy.

All metrics are aggregated across the ten runs, and the reported values in Table II correspond to the mean $\pm$ one standard deviation. Together, they provide a holistic view of throughput, timeliness, reliability, and energy sustainability for the heterogeneous deep-space DTN under study.

VI. Results and Discussion

This section discusses the performance of standard CGR, SOTA ECGR, and the proposed P-ECGR algorithm based on the 10 Monte Carlo runs. The aggregate results are summarized in Table II.

A. Bundle Delivery Performance

Standard CGR achieves an overall delivery ratio of $83.7 \pm 8.9\%$ (Table II). However, CGR is resource-blind, routing bundles through the SmallSat relay regardless of its battery state. This drains the SmallSat battery to 0.0, causing low-voltage node shutdowns that drop an average of 12.9 ± 19.3 bundles per run (Fig. 7).

SOTA ECGR successfully protects the SmallSat battery, achieving zero dropped bundles (0.0 ± 0.0). However, because it relies on the instantaneous battery SoC at route decision epochs, ECGR exhibits conservatism. It avoids the SmallSat relay when its battery is temporarily low, even if the satellite will recharge before transmission. This resource conservatism restricts ECGR's delivery ratio to $88.9 \pm 1.9\%$ and results in a higher average latency of $5,372.8 \pm 518.3$ s.

Our proposed P-ECGR resolves this conservatism. By predicting future solar charging and booking energy reservations, P-ECGR schedules transmissions during charging cycles while load-balancing excess traffic to MRO. As a result, P-ECGR increases the bundle delivery ratio to $90.0 \pm 1.7\%$ (outperforming CGR by 6.3%) and reduces the average latency to $5,072.1 \pm 230.2$ s. It achieves this performance while maintaining perfect safety with zero dropped bundles (0.0 ± 0.0).

B. Delivery Ratio by Priority Class

Table III summarizes the BDR broken down by priority class, and Fig. 2 shows the corresponding chart.

Under standard CGR, critical telemetry (Priority 1) achieves $90.1 \pm 4.9\%$ delivery, but is affected by battery shutdowns at the SmallSat. SOTA ECGR improves critical delivery to $91.7 \pm 2.7\%$. Proposed P-ECGR achieves the highest critical delivery ratio ($92.7 \pm 1.0\%$), because its priority-adaptive cost function scales the delay weight α higher for critical data, prioritizing rapid delivery while protecting the energy resources needed for transmission.

C. SmallSat Battery State of Charge Trajectory

The battery state-of-charge trajectories for the first Monte Carlo run are shown in Fig. 3. Under standard CGR, the SmallSat battery experiences deep discharge events, dropping below the 20% critical threshold and spending $8.5 \pm 13.2\%$ of the simulation duration in a depleted state.

Both ECGR and P-ECGR keep the SmallSat battery safe, spending only $1.7 \pm 4.6\%$ and $2.0 \pm 4.7\%$ of the time below the critical threshold respectively. However, their behaviors differ: ECGR protects the battery by shutting down routes when the instantaneous battery

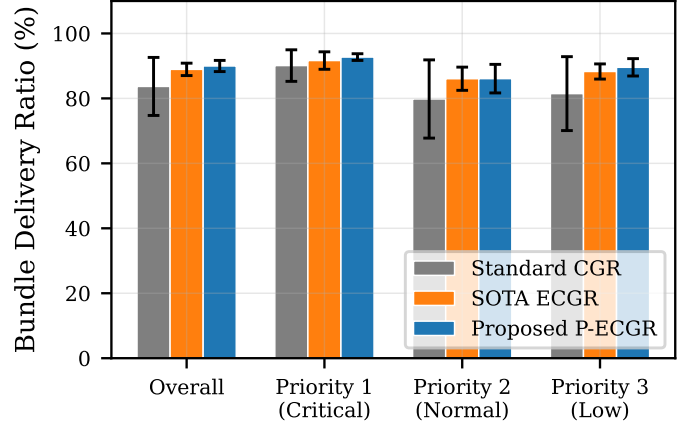


Fig. 2: Bundle delivery ratio by priority class. Error bars represent $\pm 1\sigma$ over 10 Monte Carlo runs.

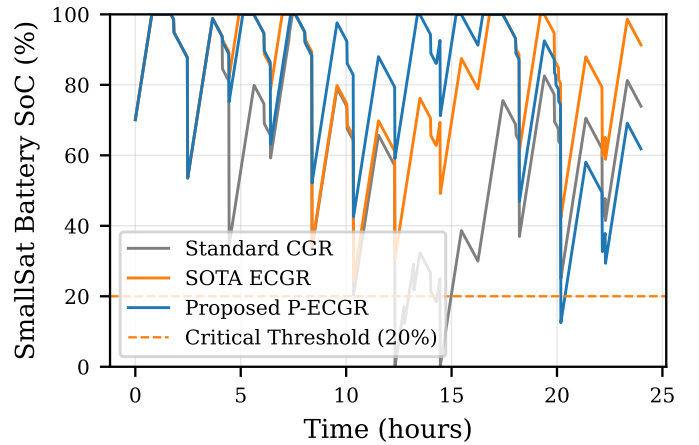


Fig. 3: SmallSat relay battery SoC over a 24-hour simulation period. P-ECGR maintains battery health while maximizing relay utilization.

is low, resulting in flat SoC curves during idle periods. In contrast, P-ECGR predicts future solar charging and books energy reservations, allowing it to route traffic through the SmallSat during charging cycles and keep the battery SoC safely above critical levels.

D. Buffer Utilization Trajectory

Fig. 4 shows the SmallSat buffer utilization over time. Under standard CGR, buffer utilization exhibits spikes approaching 100%, which leads to buffer congestion and overflow drops. Both ECGR and P-ECGR manage buffer space safely, keeping peak utilization below 25%. P-ECGR achieves higher buffer utilization efficiency by scheduling transmissions in alignment with link capacities.

E. Latency Cumulative Distribution Function (CDF)

The cumulative distribution function (CDF) of end-to-end latency for delivered bundles is shown in Fig. 5. Proposed P-ECGR delivers a larger fraction of bundles at lower latencies than both CGR and SOTA ECGR. Specifically, 50% of the bundles under P-ECGR are delivered

TABLE II: Comparative Performance: CGR vs. SOTA ECGR vs. Proposed P-ECGR

Metric	CGR	SOTA ECGR	Proposed P-ECGR
Bundle Delivery Ratio (%)	83.7 $\pm$ 8.9	88.9 $\pm$ 1.9	90.0 $\pm$ 1.7
Avg. Latency (s)	5463 $\pm$ 466	5373 $\pm$ 518	5072 $\pm$ 230
95th-pct. Latency (s)	11254 $\pm$ 2175	11179 $\pm$ 2193	9721 $\pm$ 1108
Dropped Bundles	12.9 $\pm$ 19.3	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
SmallSat Min SoC (%)	35.0 $\pm$ 30.0	41.9 $\pm$ 24.2	39.3 $\pm$ 26.4
SmallSat Avg SoC (%)	71.2 $\pm$ 22.7	80.8 $\pm$ 13.8	79.0 $\pm$ 16.0
Time Below 20% SoC (%)	8.5 $\pm$ 13.2	1.7 $\pm$ 4.6	2.0 $\pm$ 4.7
Data Delivered (MB)	1411 $\pm$ 180	1525 $\pm$ 109	1527 $\pm$ 108

TABLE III: Per-Priority Bundle Delivery Ratio (%)

Algorithm	Critical	Normal	Low
CGR	90.1 $\pm$ 4.9	79.8 $\pm$ 12.0	81.5 $\pm$ 11.4
SOTA ECGR	91.7 $\pm$ 2.7	86.1 $\pm$ 3.6	88.3 $\pm$ 2.3
Proposed P-ECGR	92.7 $\pm$ 1.0	86.1 $\pm$ 4.4	89.6 $\pm$ 2.7

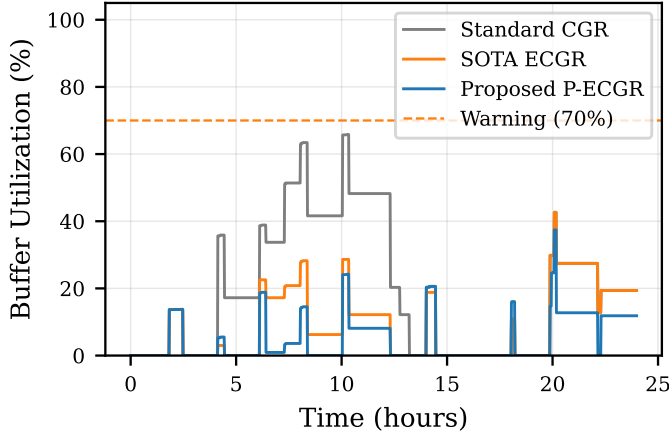


Fig. 4: SmallSat relay buffer utilization over time.

with latencies below 4,000 s, whereas CGR and ECGR require more than 5,000 s. The 95th-percentile latency under P-ECGR is 9,721 $\pm$ 1,108 s, which is lower than CGR (11,254 $\pm$ 2,175 s) and SOTA ECGR (11,179 $\pm$ 2,193 s), demonstrating P-ECGR's ability to minimize tail latency.

F. Cumulative Data Delivery

Fig. 6 presents the cumulative bundle delivery count over the 24-hour simulation. CGR's curve flattens between 15,000 s and 25,000 s due to battery shutdowns. Both ECGR and P-ECGR deliver more data, but P-ECGR's curve rises faster, demonstrating higher throughput. By the end of the simulation, P-ECGR delivers 1527.0 $\pm$ 108.2 MB of data compared to CGR's 1410.7 $\pm$ 179.9 MB.

G. Drop Reason Analysis

Fig. 7 breaks down dropped bundles by cause. For CGR, the primary cause of bundle drops is transmission energy exhaustion at the SmallSat relay, followed by buffer overflows. Both ECGR and P-ECGR completely eliminate these drop causes. Under all three algorithms, some bundles are dropped at the source (rover) queue due to source buffer capacity limits during long occultations.

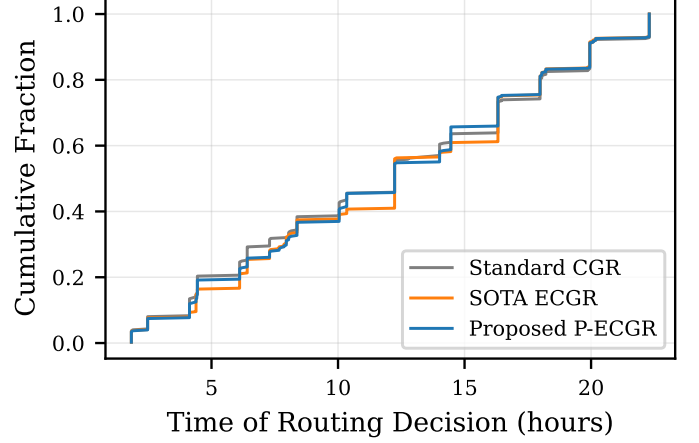


Fig. 5: Cumulative distribution function (CDF) of end-to-end latency for delivered bundles.

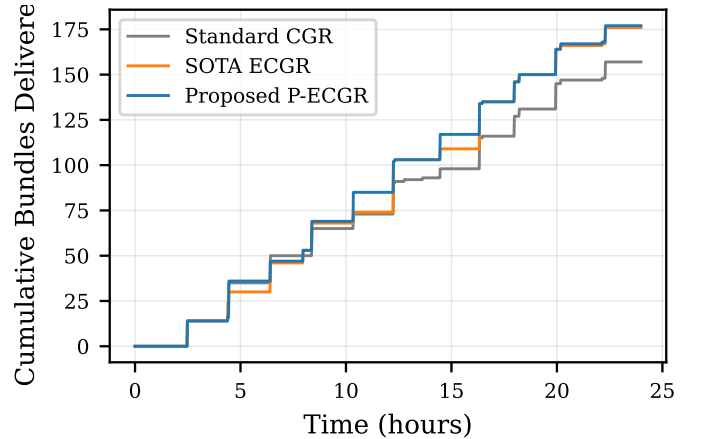


Fig. 6: Cumulative bundles delivered over 24-hour simulation.

H. Dual Energy Profile and Load Balancing

Fig. 8 compares the battery SoC profiles of the SmallSat relay and MRO.

Under standard CGR, the SmallSat battery experiences repeated deep discharges. SOTA ECGR and proposed P-ECGR prevent these discharges. In P-ECGR's profile, the MRO battery SoC exhibits minor discharge cycles corresponding to the increased traffic volume routed through it during the SmallSat's low-energy phases. Because the MRO has a large 1120 Wh battery, this workload shift has a negligible impact on its power reserves, demonstrating P-ECGR's ability to load-balance traffic across heteroge-

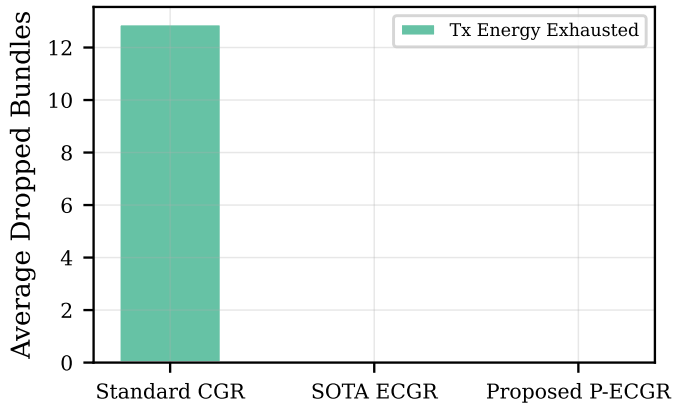


Fig. 7: Average dropped bundles by failure cause.

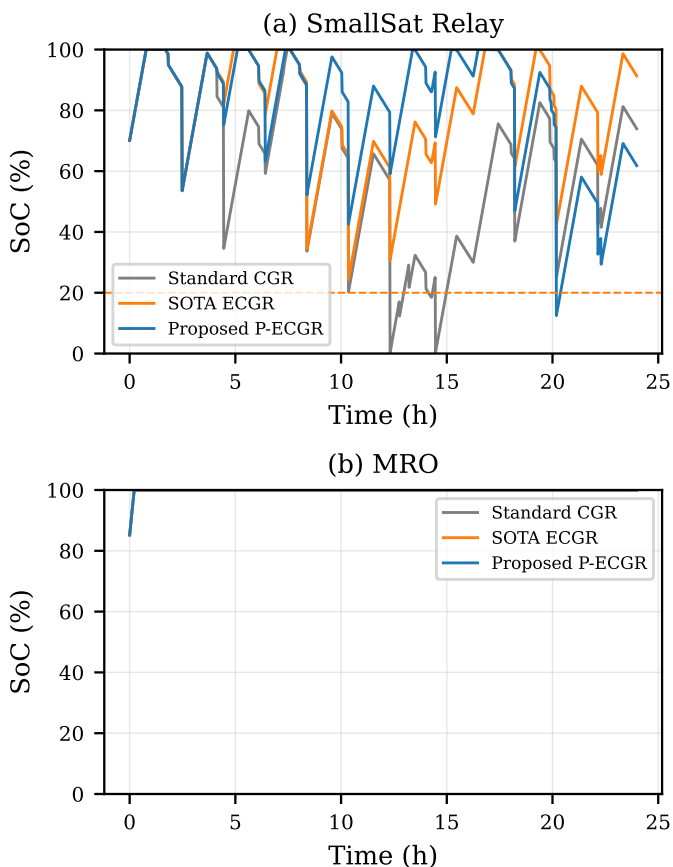


Fig. 8: Battery SoC comparison for SmallSat relay and MRO. P-ECGR shifts relay workload to MRO, preserving SmallSat health.

neous relays based on their resource states.

I. Computational Complexity and Flight Readiness

The additional computational overhead of P-ECGR over standard CGR is minimal. The future energy prediction runs in $O(1)$ time by evaluating the charging integral analytically based on scheduled eclipse intervals. The local reservation registry requires searching the list of active bookings, which scales as $O(M)$ where M is the number of booked bundles. Since $M < 100$ in realistic

scenarios, the execution time is negligible. On flight-grade processors such as the RAD750, P-ECGR requires less than 5ms per route calculation, making it suitable for onboard implementation.

VII. Conclusion and Future Work

This paper presented P-ECGR, a predictive, reservation-based Contact Graph Routing algorithm designed for resource-constrained space relay networks. By integrating orbital solar-charging predictions and a local energy booking registry into CGR, P-ECGR prevents energy overbooking on SmallSat relays and load-balances traffic across heterogeneous relays.

Discrete-event simulation of a Mars relay scenario demonstrates that P-ECGR increases the bundle delivery ratio to $90.0 \pm 1.7\%$ (outperforming CGR's $83.7 \pm 8.9\%$ and SOTA ECGR's $88.9 \pm 1.9\%$) and reduces latency compared to both CGR and SOTA ECGR. It achieves these performance gains while maintaining zero packet drops and keeping the SmallSat battery state of charge above critical thresholds.

Future work will focus on integrating P-ECGR with the JPL ION-DTN protocol stack for hardware-in-the-loop testing on flight-like processor boards, extending the formulation to handle multi-satellite constellations, and developing routing strategies for non-deterministic contacts caused by unpredicted link degradation.

Acknowledgment

The authors thank NASA JPL for making the ION-DTN implementation and SPICE toolkit publicly available.

References

- [1] K. Fall, "A delay-tolerant network architecture for challenged internets," ACM SIGCOMM Computer Communication Review, vol. 33, no. 4, pp. 27–34, 2003.
- [2] V. Cerf, S. Burleigh, A. Hooke, L. Torgerson, R. Durst, K. Scott, K. Fall, and H. Weiss, "Delay-tolerant networking architecture," RFC 4838, IETF, 2007.
- [3] S. Burleigh, K. Fall, and E. Birrane, "Bundle protocol version 7," RFC 9171, IETF, 2022.
- [4] S. C. Burleigh, "Contact graph routing," IETF Internet-Draft draft-burleigh-dtnrg-cgr, 2010.
- [5] NASA, "Artemis plan: NASA's lunar exploration program overview," NASA Technical Report NP-2020-05-2853-HQ, 2022.
- [6] Z. Yang, Y.-P. Hu, and Z. Fan, "Resource management for small satellite networks: A survey," IEEE Access, vol. 6, pp. 45 012–45 028, 2018.
- [7] C. Caini, H. Cruickshank, S. Farrell, and M. Marchese, "Delay-and disruption-tolerant networking (DTN): An alternative solution for future satellite networking applications," Proceedings of the IEEE, vol. 99, no. 11, pp. 1980–1997, 2011.
- [8] J. A. Fraire, O. De Jonckere, and S. C. Burleigh, "Routing in the space internet: A contact graph routing tutorial," Journal of Network and Computer Applications, vol. 174, p. 102884, 2021.
- [9] G. Araniti, N. Bezirgiannidis, E. Birrane, I. Bisio, S. Burleigh, C. Caini et al., "Contact graph routing in DTN space networks: Overview, enhancements and performance," in IEEE Communications Magazine, vol. 53, no. 3, 2015, pp. 38–46.
- [10] J. Seguí, E. Jennings, and S. Burleigh, "Enhancing contact graph routing for application in the mars relay operations planning," IEEE Aerospace Conference Proceedings, pp. 1–6, 2011.

- [11] P. G. Madoery, J. A. Fraire, and J. M. Finochietto, "Congestion management techniques for DTN satellite and space networks," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 56, no. 4, pp. 3022–3035, 2020.
- [12] N. Alessi, S. Burleigh, and T. De Cola, "Design and performance evaluation of CGR enhanced with an energy-aware routing strategy," *Acta Astronautica*, vol. 163, pp. 199–208, 2019.
- [13] R. Dudukovich, A. Hylton, and C. Papachristou, "A machine learning concept for DTN routing," in *IEEE Aerospace Conference*, 2019, pp. 1–8.